Employee Transfer Activity

AccessGuard Health – Healthcare IAM Operations Simulation

Mover scenario: transferring a nurse to Pharmacy and proving his access changes with his role

Project Name	AccessGuard Health
Project Type	Healthcare IAM Operations Simulation – Home Lab
Organization	Horizon Valley Medical Center (HVMC) – fictional
Activity	Activity 2 – Employee Transfer (Mover)
Scope	On-premises directory (Samba4 Active Directory)
Document Version	1.0
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1. Purpose

The **mover** is the second step of the identity lifecycle: when someone changes jobs, their old access must be taken away and the new access given, so that they never keep permissions from a role they no longer hold. This activity transfers a nurse to Pharmacy and checks the result from a client machine: his OU and group change, he can no longer write to the EHR, and he can write to the Pharmacy system.

The employee is fictional. In Microsoft Entra ID the same transfer was recorded earlier (ticket HVMC-JML-202); this activity applies it to the on-premises directory so that both directories agree.

2. Scenario

Item	Before	After
Employee	Noah Fischer (fictional), Nurse	Noah Fischer, Pharmacy Technician
Username	noah.fischer	noah.fischer
OU	OU=Nursing,OU=Users	OU=Pharmacy,OU=Users
Role group	SG-Nursing-EHR-RW	SG-Pharmacy-PharmacySystem-RW
EHR share	Read/write	Read-only
PharmacySystem share	Read-only	Read/write

The access comes from the RBAC matrix: Nursing works with the electronic health record, while Pharmacy owns the pharmacy system and only reads the EHR.

3. Step 1 – Before: Current OU and Group

The domain controller was asked for Noah's location and group membership before any change:

```
sudo samba-tool user show noah.fischer | grep -E "^dn:|memberOf"
```

```
ubuntu@hvmc-dc01:~$ sudo samba-tool user show noah.fischer | grep -E "^dn:|memberOf"
[sudo] password for ubuntu:
dn: CN=Noah Fischer,OU=Nursing,OU=Users,DC=hvmc,DC=lan
memberOf: CN=SG-Nursing-EHR-RW,OU=Groups,DC=hvmc,DC=lan
ubuntu@hvmc-dc01:~$
```

Figure 1: Before: Noah is in OU=Nursing and in SG-Nursing-EHR-RW

4. Step 2 – Set a Lab Password for Testing

A new lab password was set for Noah so that the access tests could be run as him. It follows the domain policy (minimum 12 characters, complexity) and is not shown in any screenshot.

```
sudo samba-tool user setpassword noah.fischer
```

```
ubuntu@hvmc-dc01:~$ sudo samba-tool user setpassword noah.fischer
New Password:
Retype Password:
Changed password OK
ubuntu@hvmc-dc01:~$
```

Figure 2: Password changed

5. Step 3 – Before-Test: Write to EHR as a Nurse

From the client **hvmc-client01**, Noah created a small test folder in the EHR share. Creating a folder needs write permission, so it proves he can write.

```
smbclient //hvmc-dc01.hvmc.lan/EHR -U 'HVMC\noah.fischer' -c 'mkdir nursing_test; ls'
```

```
ubuntu@hvmc-client01:~$ smbclient //hvmc-dc01.hvmc.lan/EHR -U 'HVMC\noah.fischer' -c 'mkdir nursing_test; ls'
Password for [HVMC\noah.fischer]:
  .                                   D        0  Tue Nov  3 14:53:43 2026
  ..                                  D        0  Tue Nov  3 14:53:43 2026
  nursing_test                        D        0  Tue Nov  3 14:53:43 2026

                10218772 blocks of size 1024. 4787756 blocks available
ubuntu@hvmc-client01:~$
```

Figure 3: Before: the folder nursing_test is created, so Noah can write to EHR

6. Step 4 – The Transfer

Three commands were run in order. The old group is removed **before** the new one is added, so Noah never holds both roles at once (separation of duties). Then his account was moved to the Pharmacy OU.

```
sudo samba-tool group removemembers SG-Nursing-EHR-RW noah.fischer
sudo samba-tool group addmembers SG-Pharmacy-PharmacySystem-RW noah.fischer
sudo samba-tool user move noah.fischer "OU=Pharmacy,OU=Users"
```

```
ubuntu@hvmc-dc01:~$ sudo samba-tool group removemembers SG-Nursing-EHR-RW noah.fischer
Removed members from group SG-Nursing-EHR-RW
ubuntu@hvmc-dc01:~$ sudo samba-tool group addmembers SG-Pharmacy-PharmacySystem-RW noah.fischer
Added members to group SG-Pharmacy-PharmacySystem-RW
ubuntu@hvmc-dc01:~$ sudo samba-tool user move noah.fischer "OU=Pharmacy,OU=Users"
Moved user "noah.fischer" into "OU=Pharmacy,OU=Users,DC=hvmc,DC=lan"
ubuntu@hvmc-dc01:~$
```

Figure 4: Removed from Nursing, added to Pharmacy, moved to the Pharmacy OU

7. Step 5 – After: New OU and Group

The same check as Step 1 was repeated:

```
ubuntu@hvmc-dc01:~$ sudo samba-tool user show noah.fischer | grep -E "^dn:|memberOf"
dn: CN=Noah Fischer,OU=Pharmacy,OU=Users,DC=hvmc,DC=lan
memberOf: CN=SG-Pharmacy-PharmacySystem-RW,OU=Groups,DC=hvmc,DC=lan
ubuntu@hvmc-dc01:~$
```

Figure 5: After: Noah is in OU=Pharmacy and in SG-Pharmacy-PharmacySystem-RW only

8. Step 6 – After-Test as a Pharmacy Technician

First the EHR test was repeated. Pharmacy only has read access to the EHR, so creating a folder should now be refused, while listing the folder still works:

```
smbclient //hvmc-dc01.hvmc.lan/EHR -U 'HVMC\noah.fischer' -c 'mkdir pharmacy_test; ls'
```

```
ubuntu@hvmc-client01:~$ smbclient //hvmc-dc01.hvmc.lan/EHR -U 'HVMC\noah.fischer' -c 'mkdir pharmacy_test; ls'
Password for [HVMC\noah.fischer]:
NT_STATUS_ACCESS_DENIED making remote directory \pharmacy_test
  .                                   D        0  Tue Nov  3 14:53:43 2026
  ..                                  D        0  Tue Nov  3 14:53:43 2026
  nursing_test                        D        0  Tue Nov  3 14:53:43 2026

                10218772 blocks of size 1024. 4787756 blocks available
ubuntu@hvmc-client01:~$
```

Figure 6: After: creating a folder in EHR is refused (NT_STATUS_ACCESS_DENIED); listing still works

Then the Pharmacy system share was tested. Pharmacy has read/write access there, so the folder should be created:

```
smbclient //hvmc-dc01.hvmc.lan/PharmacySystem -U 'HVMC\noah.fischer' -c 'mkdir pharmacy_test; ls'
```

```
ubuntu@hvmc-client01:~$ smbclient //hvmc-dc01.hvmc.lan/PharmacySystem -U 'HVMC\noah.fischer' -c 'mkdir pharmacy_test; ls'
Password for [HVMC\noah.fischer]:
  .                                   D        0  Tue Nov  3 15:02:13 2026
  ..                                  D        0  Tue Nov  3 15:02:13 2026
  pharmacy_test                       D        0  Tue Nov  3 15:02:13 2026

                10218772 blocks of size 1024. 4787744 blocks available
ubuntu@hvmc-client01:~$
```

Figure 7: After: the folder pharmacy_test is created, so Noah can write to PharmacySystem

9. Results

Check	Expected	Result
OU before the transfer	OU=Nursing	OU=Nursing
Group before the transfer	SG-Nursing-EHR-RW	SG-Nursing-EHR-RW
Write to EHR before	Allowed	Allowed (nursing_test created)
OU after the transfer	OU=Pharmacy	OU=Pharmacy
Group after the transfer	SG-Pharmacy-PharmacySystem-RW only	SG-Pharmacy-PharmacySystem-RW only
Write to EHR after	Refused	Refused: NT_STATUS_ACCESS_DENIED
Read EHR after	Allowed	Allowed (folder listed)
Write to PharmacySystem after	Allowed	Allowed (pharmacy_test created)

10. Security Considerations

- All names and users are **fictional**; no real employee or patient data is used.
- **Remove before add:** the old role's group is removed first, so a mover never holds two roles at once.
- **Least privilege:** after the move Noah can read the EHR but can no longer change patient records.
- **No leftover access:** his old group is gone, and the test proved the old write permission no longer works.
- Access comes from groups, so the same three steps work for any transfer and can be reversed.
- The lab password is practice-only and does not appear in the screenshots.

11. Key Outcomes

- Transferred a nurse to Pharmacy on the on-premises directory, matching the Entra ID record (HVMC-JML-202).
- Showed the before and after state for his OU and group, with evidence at each step.
- Proved the change from a real client: EHR changed from write to read-only, and PharmacySystem became writable.

12. Deliverables

- Employee Transfer Activity: Lab Documentation for GitHub Portfolio (this document).
- Screenshots: before and after OU and group, the transfer commands, and the EHR and PharmacySystem access tests.